Math Workshop for Political Science
The Ohio State University
Syllabus: Summer 2026

Instructor

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Class location: 2130 Derby Hall
Class time: M–F, 9:00–11:30 EDT

Teaching Assistants

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Description

The purpose of this workshop is to provide incoming first year Ph.D. students with some fundamental skills in various mathematical techniques that are used in political science, regardless of sub-specialty, and generally to prepare students for the first-year methods sequence. The workshop is also open to continuing students, who feel that they would gain from participating in the course.

This year the course will begin on Monday, August 10th and run every weekday until Friday, August 21st. Sessions are tenantively scheduled to run from 9:00—11:30 each day. In the past, there has also been an additional meeting on Saturday. We hope this will not be necessary this year, though if we fall behind in the material, we may need to reconsider.

Textbook

Moore, W.H. and Siegel, D.A., 2013. *A Mathematics Course for Political and Social Research*. Princeton University Press. `https://press.princeton.edu/books/paperback/9780691159171/a-mathematics-course-for-political-and-social-research` (SM)

Wickham, H. and Grolemund, G., 2017. *R for Data Science*. O'Reilly. `https://r4ds.had.co.nz/` (WG)

General resources

- Preparation for the Ph.D, a guide put together by PRISM: https://polisci.osu.edu/sites/default/files/2024-05/Preparation%20for%20the%20PhD.pdf
- Another popular, but more technical, textbook is: Simon, Carl P. and Lawrence Blume., 1994, *Mathematics for Economists*. Vol. 7. New York: Norton. https://wwnorton.com/books/Mathematics-for-Economists/
- MIT Open Courseware (Mathematics): http://ocw.mit.edu/courses/#mathematics
- Khan Academy: http://www.khanacademy.org/
- MathTV: http://www.mathtv.com/videos_by_topic

Class Format

The workshop will be taught in a lecture format. During each day's session we will review the problem set from the previous day before covering the current day's material. Before each session, it is expected that students will complete the previous day's problem set and review the current day's course materials. While the course will be presented as a lecture, student participation is strongly encouraged. For each module, links to a series of short videos, lecture notes, and a daily problem set will be provided.

The day's preparatory videos are available in the course's Github repository here. Problem sets and lecture notes will be posted immediately following the end of that day's session.

There are nine problem sets, each of which will be due the following morning at the start of class. Students should plan to spend 2–4 hours each day completing these assignments and preparing for the following session.

R Introduction

The compressed nature of this class makes it impossible to give students a comprehensive introduction to all of the tools they will need in the first year methods sequence. However, a brief introduction to R and computational social science will be provided during the last few lectures. The lecture notes also contain some examples of using R to solve or visualize problems we will encounter during the course.

Our R resource for these lectures will be a notebook that we will work through together. While the Wickham and Grolemund book is a fantastic resource for one's methods coursework, it is also quite thorough and extensive. For a brief introduction and overview of R this workshop will be helpful.

Class schedule

The topics and subtopics covered in this course are provided below. The schedule is tentative and will be adjusted as we make our way through the materials.

Module 1: Notation, Definitions, and Basic Mathematics (Part I) [Day 1]

- Definition of a variable and real number systems
- Set notation and relationships
- Definition of independent and dependent variables
- Discussion of interval notation
- Definitions of types of functions
- Commutative, associative, and distributive laws
- Concepts of inequality and absolute value
- Exponent rules

Module 1: Basic Mathematics (Part II) [Day 2]

- Summation and product operators
- Solving equations, inequalities, and for roots
  - Single and multiple variables
  - Quadratic formula
  - Factoring
- Logarithms and rules

Module 2: Introduction to Probability and Statistics [Day 3]

- Factorials, permutations, and combinations
- Random experiments, sample spaces, and events
- Probability axioms and general addition rule
- Conditional probability and independence
- Law of total probability and Bayes' rule
- Random variables (discrete vs. continuous)
- PMFs, PDFs, and CDFs
- Expected value, mean, and variance (properties and calculations)

Module 3: Linear Algebra (Part I) [Day 4]

- Linear equations and linear systems
- Method of elimination

- Definition of matrices and vectors
- Matrix operators and transposes
- Dot product and matrix multiplication
- Matrix representation of systems of equations

Module 3: Linear Algebra (Part II) [Day 5]

- Linear dependence/independence
- Properties of matrix operators and symmetric matrices
- Definition of identity, zero, and idempotent matrices
- Reduced row/row echelon form (Gauss-Jordan reduction)
- Inverses and conditions for nonsingularity
- Matrix rank, determinants, and trace

Module 4: Limits and Differentiation [Day 6]

- Limits and sequences
- The difference quotient and the derivative
- Rules of differentiation for a function of one variable
- Product, quotient, and chain rules

Module 4: Advanced Differentiation and Optimization [Day 7]

- Derivative of exponential and log functions
- Implicit differentiation
- Partial differentiation, gradient, and the Hessian matrix
- Unconstrained optimization (first- and second-order conditions)
- Constrained optimization and Lagrange multipliers

Module 5: Calculus Integration + R Introduction [Day 8]

- Indefinite integrals and antidifferentiation
- Areas and Riemann sums
- Definite integrals and the Fundamental Theorem of Calculus
- Rules for definite integrals
- **R Introduction**
  - Installing R and RStudio
  - Scripts vs. terminal vs. notebooks
  - Coding best practices

Module 5: Advanced Integration Methods + R Basics [Day 9]

- Integration by substitution
- Integration by parts
- Improper integrals
- Double integrals
- **R Basics**
    - Object types
    - Data storage
    - Functional programming
    - Data visualization

Post-test and R Data Analysis [Day 10]

- Post-test exam
- **Data Analysis in R**
    - Exploratory data analysis on real data